

SAT Unit 4.2

Line, Angel, and Triangle

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Overview

- 1 Line
- 2 Angle
- 3 Triangle
- 4 Practices

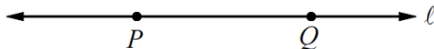
Line

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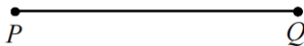
Definition of Line, Segment, and Ray

- **Line:** A straight arrangement of points extending infinitely in both directions.
- **Segment:** A part of a line with two endpoints.
- **Ray:** A part of a line with one endpoint extending infinitely in one direction.

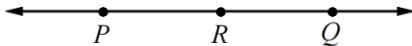
Written as: line ℓ , line PQ , or $\overleftrightarrow{PQ}$.



Written as: segment PQ , or $\overline{PQ}$.



Written as: ray PQ or $\overrightarrow{PQ}$.



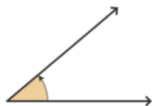


Angle

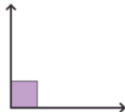
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Types of Angles

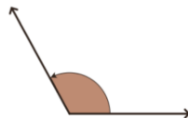
- **Acute Angle:** Less than 90° .
- **Right Angle:** Exactly 90° .
- **Obtuse Angle:** Greater than 90° but less than 180° .
- **Straight Angle:** Exactly 180° .
- **Reflex Angle:** Greater than 180° but less than 360° .
- **Complete Angle:** Exactly 360° .



ACUTE ANGLE



RIGHT ANGLE



OBTUSE ANGLE



STRAIGHT ANGLE



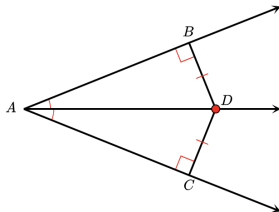
REFLEX ANGLE



COMPLETE ANGLE

Angle Bisector Theorem

Statement: If a point lies on the **bisector of an angle**, then it is equidistant from the two sides of the angle.



Mathematical Representation:

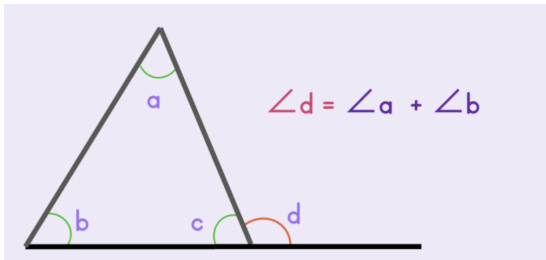
$$\left\{ \begin{array}{l} \overrightarrow{AD} \text{ bisects } \angle BAC \\ DB \perp \overrightarrow{AB} \\ DC \perp \overrightarrow{AC} \end{array} \right\} \Rightarrow DB = DC$$

Exterior Angle Theorem

Theorem Statement: In a triangle, the measure of an **exterior angle** is equal to the sum of the measures of the two **remote interior angles**.

Visual Representation: If $\angle d$ is the exterior angle, then:

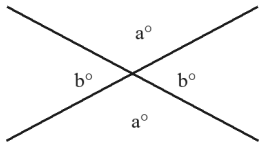
$$\angle d = \angle a + \angle b$$



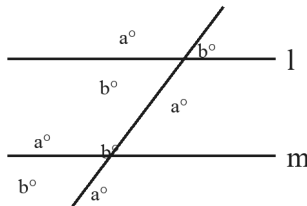
SAT Tip: This theorem is frequently used in problems involving triangles and polygon extensions. It's a shortcut to avoid summing all three triangle angles when only two are needed!

Intersecting Lines and Parallel Lines

- **Intersecting Lines:** Opposite angles are equal. Also, each pair of angles along the same line adds to 180° . In the figure above, $a + b = 180^\circ$.
- **Parallel Lines:** Eight angles are formed when a line crosses two parallel lines. The four large angles (a) are equal, and the four small angles (b) are equal.



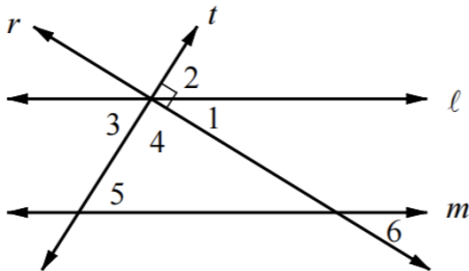
Intersecting Lines



Parallel Lines ($l \parallel m$)

Question: Finding Angle Measures

In the figure below, $\ell \parallel m$, $r \perp t$, and $m\angle 1 = 32^\circ$. Lines ℓ , r , and t intersect at one point. Find $m\angle 2$, $m\angle 3$, $m\angle 4$, and $m\angle 5$.



Answer Explanation: Finding Angle Measures

Solution:

- A right angle measures 90° .
- Using $m\angle 1 + m\angle 2 = 90^\circ$:

$$32^\circ + m\angle 2 = 90^\circ$$

$$m\angle 2 = 58^\circ$$

- Vertical angles are congruent:

$$m\angle 2 = m\angle 3 = 58^\circ$$

- A straight angle measures 180° , so:

$$m\angle 1 + m\angle 4 + m\angle 3 = 180^\circ$$

$$32^\circ + m\angle 4 + 58^\circ = 180^\circ$$

$$m\angle 4 = 90^\circ$$

- Alternate interior angles are congruent:

Answer Explanation: Finding Angle Measures

- Corresponding angles are congruent:

$$m\angle 1 = m\angle 6 = 32^\circ$$

Final Answers:

$$m\angle 2 = 58^\circ, \quad m\angle 3 = 58^\circ, \quad m\angle 4 = 90^\circ, \quad m\angle 5 = 58^\circ.$$

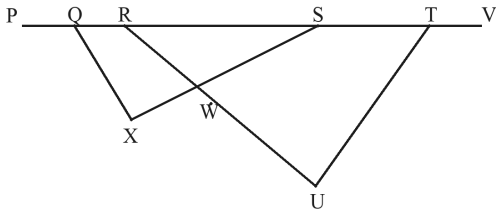
Question: Finding $\angle TUR$

In the figure shown, points $Q, R, S,$ and T lie on line segment PV , and line segment RU intersects line segment SX at point W .

Given:

- $\angle SQX = 48^\circ$
- $\angle SXQ = 86^\circ$
- $\angle SWU = 85^\circ$
- $\angle VTU = 162^\circ$

What is the measure, in degrees, of $\angle TUR$?



Answer Explanation: Finding $\angle TUR$

Solution:

- Compute $\angle XQS$ using the given angles:

$$\angle XQS = 180^\circ - \angle SXQ = 180^\circ - 86^\circ = 94^\circ$$

- Compute $\angle XSW$ using $\angle SQX$ and $\angle XQS$:

$$\angle XSW = 180^\circ - (\angle SQX + \angle XQS) = 180^\circ - (48^\circ + 94^\circ) = 38^\circ$$

- Use $\angle SWU = 85^\circ$ to find $\angle WUR$:

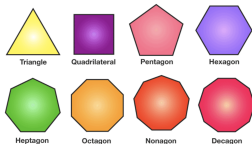
$$\angle WUR = 180^\circ - (\angle SWU + \angle XSW) = 180^\circ - (85^\circ + 38^\circ) = 57^\circ$$

- Compute $\angle TUR$ using $\angle VTU = 162^\circ$:

$$\angle TUR = 180^\circ - \angle WUR = 180^\circ - 57^\circ = 123^\circ$$

Final Answer: 123°.

Polygons and Interior Angles



Polygon Angle Sum Formula:

$$\text{Sum of interior angles} = 180^\circ(n - 2)$$

- n = number of sides
- Each additional side increases the angle sum by 180°

Regular Polygon:

All sides and angles are equal.

Measure of Each Interior Angle in a Regular Polygon:

$$\text{Each angle} = \frac{180(n - 2)}{n}$$

Question: Polygon Angle Problem

A regular polygon has an interior angle that is 4 times the exterior plus 60 the measure of its exterior angle.

Question: How many sides does the polygon have?

Choices:

- A. 9
- B. 10
- C. 12
- D. 15

Answer and Explanation

"A regular polygon has an interior angle that is 60 more than four times its exterior angle."

Then:

$$x = 4y + 60, \quad x + y = 180 \Rightarrow 4y + 60 + y = 180$$

Correct Answer: D. 15

Triangle

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Isosceles and Equilateral Triangles

Isosceles Triangle:

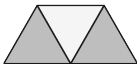
- Has two equal sides.
- The angles opposite the equal sides are also equal.
- Often appears in SAT triangle congruence or symmetry problems.

Equilateral Triangle:

- All sides are equal.
- Each angle measures 60° .
- Always a regular polygon.



Question: Area of Equilateral Triangles



A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles.

If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

- A. $2\sqrt{3}$
- B. $4\sqrt{3}$
- C. $8\sqrt{3}$
- D. 16

Answer: Area of Equilateral Triangles in a Logo

In the diagram, the trapezoid has **4 edges**, and each triangle has 3 sides. The full figure has:

$$\text{Perimeter} = 5s \Rightarrow 5s = 20 \Rightarrow s = 4$$

Area of one triangle:

$$A = \frac{\sqrt{3}}{4}(4)^2 = \frac{\sqrt{3}}{4}(16) = 4\sqrt{3}$$

Two shaded triangles:

$$2 \cdot 4\sqrt{3} = 8\sqrt{3}$$

Correct Answer: $C. 8\sqrt{3}$

Right Triangles

Definition: A triangle with one 90° angle.

Pythagorean Theorem:

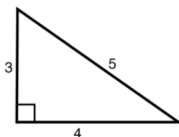
$$a^2 + b^2 = c^2$$

Where a and b are legs and c is the hypotenuse.

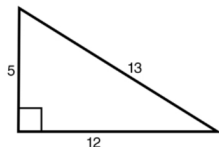
Pythagorean Triples (SAT favorites):

- (3, 4, 5)
- (5, 12, 13)

3-4-5 Triangle



5-12-13 Triangle

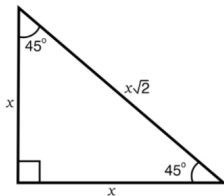


Special Right Triangles

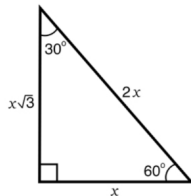
1. 45°-45°-90° Triangle

- Legs are equal.
- Hypotenuse: $x\sqrt{2}$ if each leg is x .

45-45-90 Triangle



30-60-90 Triangle



2. 30°-60°-90° Triangle

- Short leg: x
- Long leg: $x\sqrt{3}$
- Hypotenuse: $2x$

Question: Geometry – Right Triangle with Altitude

In triangle $\triangle ABC$, the measure of angle B is 90° , and $\overline{BD}$ is an altitude of the triangle.

- The length of $\overline{AB}$ is 15,
- The length of $\overline{AC}$ is 23 greater than the length of $\overline{AB}$,

What is the value of $\frac{BC}{BD}$?

- A. $\frac{15}{38}$
- B. $\frac{15}{23}$
- C. $\frac{23}{15}$
- D. $\frac{38}{15}$

Answer: Geometry – Right Triangle

We are given:

- $AB = 15$
- $AC = AB + 23 = 15 + 23 = 38$
- $\angle B = 90^\circ$, so triangle $\triangle ABC$ is a right triangle.

Using the Pythagorean Theorem:

$$AB^2 + BC^2 = AC^2 \Rightarrow 15^2 + BC^2 = 38^2$$

Correct Answer: D. $\frac{38}{15}$

Congruent Triangles: Definition and Proof

Definition: Two triangles are **congruent** if all corresponding sides and angles are equal. That means:

$$\text{Side} = \text{Side}, \quad \text{Angle} = \text{Angle}$$

$$\triangle ABC \cong \triangle DEF \Rightarrow AB = DE, \angle A = \angle D, \text{ etc.}$$

Methods to Prove Triangle Congruence:

- **SSS (Side-Side-Side):** All three sides of one triangle are equal to the corresponding sides of another.
- **SAS (Side-Angle-Side):** Two sides and the included angle are congruent.
- **ASA (Angle-Side-Angle):** Two angles and the included side are congruent.
- **HL (Hypotenuse-Leg):** In right triangles, hypotenuse and one leg are congruent.

SAT Tip: Do **not** assume triangle congruence from AAA or SSA — these are not valid for proving congruence.

Question: Triangle Congruence

In triangles ABC and DEF , angles B and E each have measure 27° , and angles C and F each have measure 41° .

Which additional piece of information is sufficient to determine whether triangle ABC is congruent to triangle DEF ?

- A. The measure of angle A
- B. The length of side AB
- C. The lengths of sides BC and EF
- D. No additional information is necessary

Answer: Triangle Congruence

Since two angles are known in each triangle:

$$\angle B = \angle E = 27^\circ, \quad \angle C = \angle F = 41^\circ$$

We can determine the third angle because the angles of a triangle sum to 180° :

$$\angle A = \angle D = 180^\circ - 27^\circ - 41^\circ = 112^\circ$$

Therefore, all three angles in both triangles are known and equal:

$$\angle A = \angle D, \quad \angle B = \angle E, \quad \angle C = \angle F$$

Thus, the two triangles are similar (AAA), but to determine **congruence**, one side must be known.

If one side length (e.g., AB) is known, the Side-Angle-Angle (ASA) or Angle-Side-Angle (ASA) congruence criterion can be applied.

Answer: C

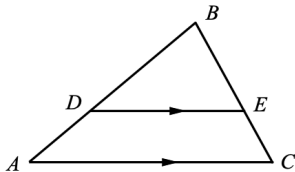
Similar Triangles

Definition: Triangles are similar if their corresponding angles are equal and corresponding sides are in proportion.

Symbolically: $\triangle BAC \sim \triangle BDE$

Properties:

- All corresponding angles are equal.
- All corresponding sides are proportional.
- $\frac{DB}{AB} = \frac{BE}{BC} = \frac{DE}{AC}$



SAT Tip: Use proportions to solve for missing side lengths or establish ratios between parts of a figure.

Similar Triangles: Definition and Proof Methods

Definition: Two triangles are **similar** if their corresponding angles are equal and their corresponding sides are in proportion.

$$\triangle ABC \sim \triangle DEF \Rightarrow \angle A = \angle D, \angle B = \angle E, \angle C = \angle F$$

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$$

Ways to Prove Similar Triangles:

- **AA (Angle-Angle):** If two angles of one triangle are equal to two angles of another, the triangles are similar.
- **SSS (Side-Side-Side) Similarity:** If all three pairs of corresponding sides are in proportion.
- **SAS (Side-Angle-Side) Similarity:** If two sides are in proportion and the included angle is equal.

Question: Triangle Similarity Criterion

In triangles LMN and RST , angles L and R each have measure 60° , $LN = 10$, and $RT = 30$.

Which additional piece of information is sufficient to prove that triangle LMN is similar to triangle RST ?

- A. $MN = 7$ and $ST = 7$
- B. $MN = 7$ and $ST = 21$
- C. The measures of angles M and S are 70° and 60° , respectively
- D. The measures of angles M and T are 70° and 50° , respectively

Answer: Triangle Similarity Criterion

Answer:

Similar Triangles in Right Triangle Model

Concept: In a right triangle, when an altitude is drawn from the right angle to the hypotenuse, it creates two smaller triangles that are similar to the original triangle and to each other.

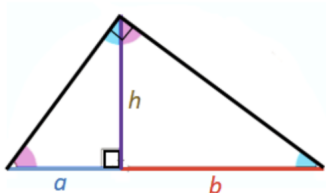
Proportional Relationship:

$$\frac{a}{h} = \frac{h}{b} \Rightarrow h = \sqrt{ab}$$

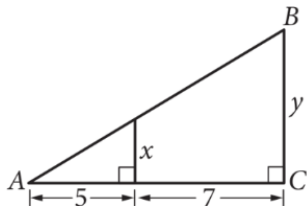
where:

- a and b are the segments of the hypotenuse,
- h is the altitude to the hypotenuse.

Tip: This model leads to fast solutions when given two sides and asked for height or area.



Question: Triangle Area and Geometry Constraints



Given:

- The area of triangle ABC is at least 48 but no more than 60.
- y is an integer.

What is one possible value of x ?

Answer: Triangle Area and Geometry Constraints

We are given:

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 12 \times y = 6y$$

$$48 \leq 6y \leq 60 \Rightarrow 8 \leq y \leq 10$$

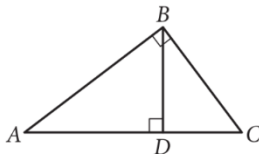
So possible integer values for y are 8, 9, or 10.

We are told that x is the distance from point A to where the perpendicular from B hits base AC , which lies between A and C . Since $AC = 12$, one possible value of x that lies between 0 and 12 is:

$$\frac{25}{6}$$

Possible Answer: 5

Question: Right Triangle Segment Length



In the figure above:

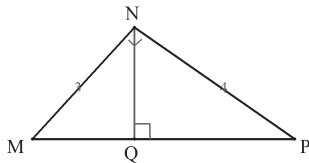
- $BD = 6$
- $AD = 8$

What is the length of $\overline{DC}$?

Answer: Right Triangle Segment Length

Answer: 4.5

Question: Right Triangle Geometry



In the figure above, what is the length of segment $\overline{NQ}$?

- A. 2.2
- B. 2.3
- C. 2.4
- D. 2.5

Answer: Right Triangle Geometry

Use the full triangle $\triangle MNP$ with legs $MN = 3$ and $NP = 4$. That makes MP the hypotenuse:

$$MP^2 = MN^2 + NP^2 = 3^2 + 4^2 = 9 + 16 = 25 \Rightarrow MP = 5$$

So triangle $\triangle MNP$ is a right triangle with hypotenuse 5. Since $\angle MQN$ is a right angle, NQ is the height to the hypotenuse.

Area of triangle using base and height:

$$\text{Area} = \frac{1}{2} \cdot MP \cdot NQ = \frac{1}{2} \cdot 5 \cdot NQ$$

Area using two legs:

$$\text{Area} = \frac{1}{2} \cdot 3 \cdot 4 = 6$$

Equating both:

$$\frac{1}{2} \cdot 5 \cdot NQ = 6 \Rightarrow \frac{5}{2} \cdot NQ = 6 \Rightarrow NQ = \frac{12}{5} = 2.4$$

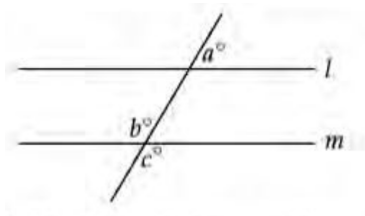
Correct Answer: C. 2.4

Practices

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Transversal Angles and Parallel Lines

Question: In the figure below, lines l and m are parallel.



If $b = 9k + 15$ and $c = 16k - 76$, what is the value of a ?

Solution

Solution:

Answer: $a = 48^\circ$.

Area of an Isosceles Right Triangle

Question: The hypotenuse of an isosceles right triangle has a length of 12 units. What is the area of the triangle, in square units?

A $18\sqrt{2}$

B 36

C 48

D $36\sqrt{2}$

Solution

Solution:

- Let the length of each leg of the isosceles right triangle be x .
- Then:

$$x^2 + x^2 = 12^2$$

$$2x^2 = 144$$

$$x^2 = 72$$

$$x = 6\sqrt{2}$$

- Therefore, the area is:

$$A = \frac{1}{2} \times x \times x = \frac{1}{2} \times (6\sqrt{2}) \times (6\sqrt{2}) = \frac{1}{2} \times 72 = 36$$

Answer: 36

Similarity of Triangles

Question: In triangles LMN and RST , angles L and R each have measure 127° . Which additional piece of information is sufficient to determine whether triangle LMN is similar to triangle RST ? The measures of angles N and T . The lengths of sides LM and RS . The measures of angles S and T . The lengths of sides MN and ST .

Solution

Solution: To determine whether two triangles are similar, we need:

- Two pairs of corresponding angles to be congruent (AA criterion).
- Alternatively, corresponding sides to be proportional (SSS or SAS criterion).

Given $\angle L = \angle R = 127^\circ$:

- Knowing either $\angle N$ and $\angle T$ or $\angle S$ and $\angle T$ would be sufficient by AA criterion.

Therefore:

Option A: The measures of angles N and T

Similarity of Triangles

Question: In triangles LMN and RST , angles L and R each have measure 127° . Which additional piece of information is sufficient to determine whether triangle LMN is similar to triangle RST ?

- A The measures of angles N and T .
- B The lengths of sides LM and RS .
- C The measures of angles S and T .
- D The lengths of sides MN and ST .

Solution

Solution: To determine whether two triangles are similar, we need:

- Two pairs of corresponding angles to be congruent (AA criterion).
- Alternatively, corresponding sides to be proportional (SSS or SAS criterion).

Given $\angle L = \angle R = 127^\circ$:

- Knowing either $\angle N$ and $\angle T$ or $\angle S$ and $\angle T$ would be sufficient by AA criterion.

Therefore:

Option A: The measures of angles N and T

Congruence of Triangles

Question: In triangles LMN and PQR , angles M and Q each have measure 54° , $MN = 16$, and $QR = 16$. Which additional piece of information is sufficient to prove that triangle LMN is congruent to triangle PQR ?

Answer Choices:

- A Angles L and R each have measure 44° .
- B Angles N and P each have measure 44° .
- C $LM = 11$ and $PQ = 11$.
- D $LN = 11$ and $PR = 11$.

Question

C

Thank You!

We appreciate your time and focus.

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